

Capstone Project

Online Retail

Intellipaat

Executive-Post-Graduate-Certification-in-Data-Science-and-Artificial-Intelligence

Date- Oct 2024



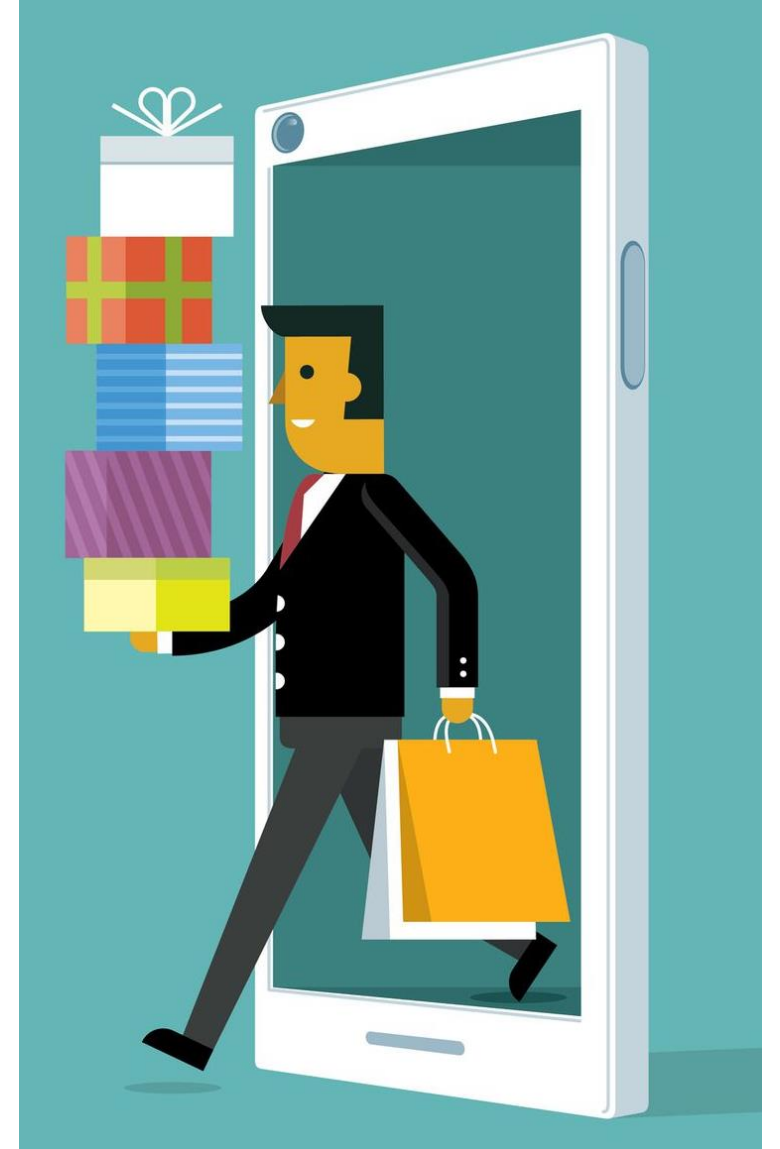
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The Problem Statement

An **online retail store** is trying to understand the various customer purchase patterns for their firm.

It has requested Intellipaat to come up with useful insights on customer behavior and create customer groups.



Project Objective

Pre-Preprocessing & Cleaning

- Shape of data
- Data type of columns
- Identify Null values
- Handle Duplicates values

Exploratory Data Analysis

- Finding Top 5 and bottom 5
 - Spending value
 - Frequency of visits
 - Dates by sales value
 - Items by sales quantity
- Number of Countries

Predictive models

- Understanding and analyzing data using RFM model
- Using K means machine learning model
- Compare the outputs of both the models

Pre Processing

Pre- processing

- **Dataset Overview:**

- **541909** rows
- **8** columns

- **Data Types:**

- **InvoiceNo**: Invoice number.[**object**]
- **StockCode**: Product (item) code.[**object**]
- **Description**: Product (item) name. [**object**]
- **Quantity**: The quantities of each product (item) per transaction. [**int64**]
- **InvoiceDate**: Invoice Date and time. The day and time when each transaction was generated. [**object**]
- **UnitPrice**: Unit price. Product price per unit in sterling.[**float64**]
- **CustomerID**: Customer number. [**Float64**]
- **Country**: Country name. the name of the country where each customer resides.[**object**]

Pre- processing

- **Null and Duplicates:**
 - There are **136534 NULL** values in the **Dataset**.
 - **135080 NULL** in the **CustomerID** Column
 - **1454 NULL** in the **Description** Column
 - Percentage of **CustomerID** missing: **24.93 %**
 - Percentage of **Product Descriptions** missing: **0.27 %**
- Having large number of NULL in Customer ID may create problem- hence, Instead of deleting such a large group- we have **assigned InvoiceNo as the CustomerID** wherever CustomerID is missing.
- After clearing the NULL values
 - **Number of transactions** in the dataset : 25900
 - **Number of different products** bought in the dataset : 4070
 - **Number of unique customers** in the dataset : 8082
 - **Number of countries** in the dataset: 38

Exploratory Data Analysis

Exploratory Data Analysis

- Top 5 customers by total spending amount.

CustomerID	Unit Price
14096.0	41376.33
15098.0	40278.90
14911.0	31060.66
12744.0	25108.89
16029.0	24111.14

- Bottom 5 customers by total spending amount.

CustomerID	UnitPrice
A563186	-11062.06
A563187	-11062.06
564927	0.00
543258	0.00
543259	0.00

Exploratory Data Analysis

- Top 5 customer by number of Invoices

CustomerID	InvoiceNo
14911	248
12748	224
17841	169
14606	128
15311	118

- Bottom 5 customer by number of Invoices

CustomerID	InvoiceNo
C580605	1
16881	1
551326	1
551327	1
551328	1

Exploratory Data Analysis

- Top 5 dates of maximum sales

InvoiceDate	Invoice_Value
11/14/2011	112141.11
9/20/2011	109286.21
12/8/2011	81417.78
11/23/2011	78480.7
10/5/2011	75244.43

- Top 5 dates of minimum sales

InvoiceDate	Invoice_Value
1/5/2011	-1566.23
2/6/2011	3457.11
3/13/2011	4137.62
8/14/2011	5150.18
2/13/2011	5535.4

Exploratory Data Analysis

- Top 5 items by Sales Quantity

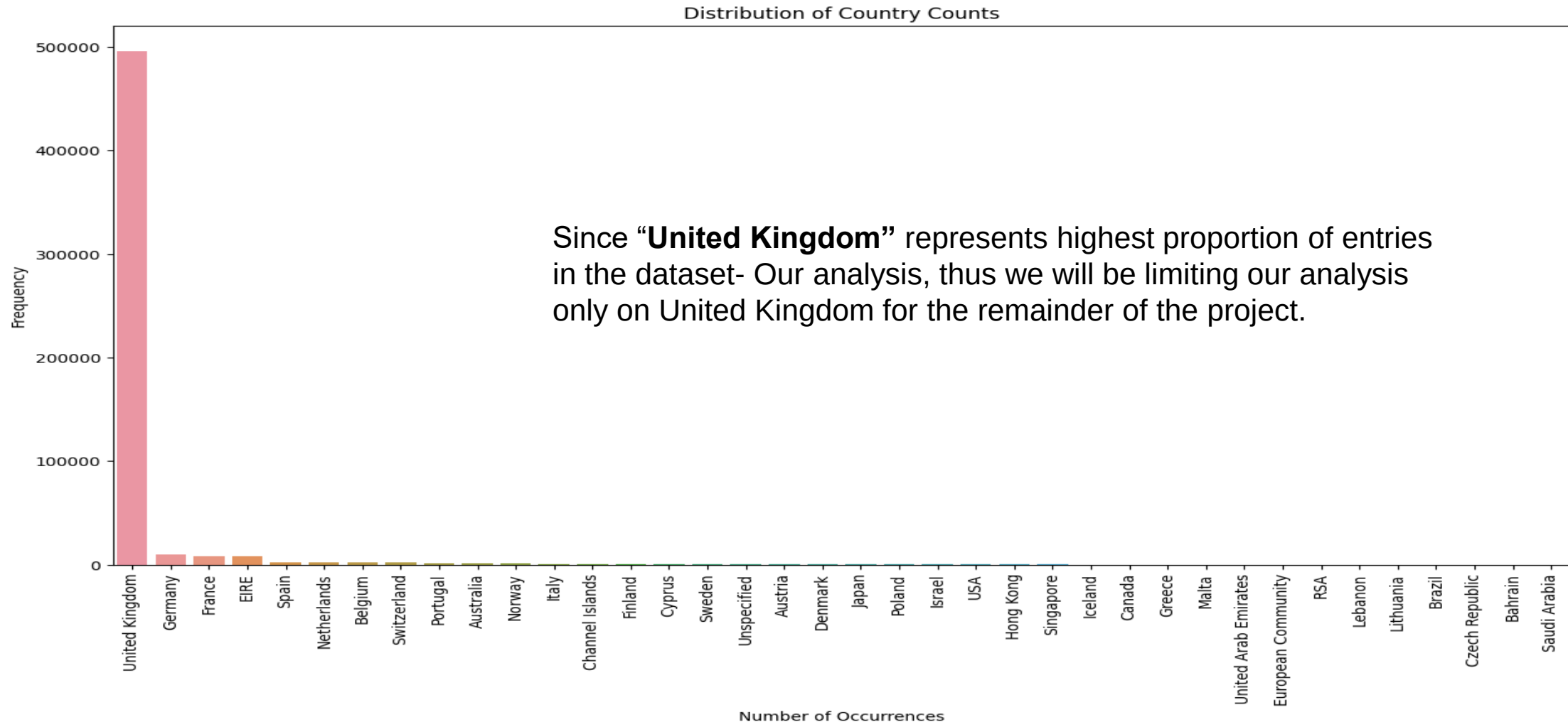
StockCode	Description	Quantity
2997	WORLD WAR 2 GLIDERS ASSTD DESIGNS	53847
3515	JUMBO BAG RED RETROSPOT	47363
3305	ASSORTED COLOUR BIRD ORNAMENT	36381
1147	POPCORN HOLDER	36334
454	PACK OF 72 RETROSPOT CAKE CASES	36039

- Bottom 5 items by Sales Quantity

StockCode	Description	Quantity
1907	printing smudges/thrown away	-19200
1905	Printing smudges/thrown away	-9058
2821	throw away	-5368
2931	Unsaleable, destroyed.	-4830
173	wrongly marked. 23343 in box	-3100

Exploratory Data Analysis

Distribution of Number of countries in the dataset.



Prediction Models

Motivation for Choosing a Model

An important marketing strategy e-commerce businesses use for analyzing and predicting customer value is customer segmentation.

Such Customer segmentation can –

1. Recognize who are our most valuable customers
2. Increase revenue
3. Increase customer retention
4. Learn more about the trends and behaviors of our customers
5. Define customers that are at risk

We will Start with **RFM analysis** and then support it with **K Means model**

RFM Model- explained !

RFM stands for **Recency, Frequency, Monetary Value (RFM)**.

- **Recency : Days since last purchase**
- **Frequency : Number of past purchases**
- **Monetary Value : Total amount of money spent**

This model is very useful for segmenting customers based on past buying behavior. It answers the following questions:

1. Who are our **best customers**?
2. Who has the **potential to be converted into more profitable** customers?
3. Which customers do we need to **retain**?
4. Which group of **customers is most likely to respond** to our marketing campaign?

RFM Model- explained !

Once we segment customers on the parameters of Recency, Frequency and Monetary Value then we can define “Customer Segment” (such as Champions, Loyal Customers etc.)

This will help digital marketing managers to define customized actions for each group

Recency	Frequency	Monetary value	Customer Segment	Activity	Actions
H	H	H	Champions	Bought recently, buy often, and spend the most!	Reward them. Can be early adopters for new products. Will promote your brand.
H	H	L	Loyal Customers	visited often, Responsive to promotions. Makes small purchases.	Upsell higher-value products. Ask for reviews. Engage them.
H	L	H	At Risk	Spent big money and recently. Frequency of visits is less	Send personalized emails to reconnect, offer renewals, and provide helpful resources.
H	L	L	Recent Customers	Bought most recently, but not often	Provide onboarding support, give them early success, and start building relationships.
L	H	H	Can't Lose Them	Made biggest purchases, and often. But haven't returned for a long time.	Win them back via renewals or newer products, don't lose them to competition, talk to them.
L	H	L	Promising	Have shopped recently. Shopped less value earlier but repeatedly.	Create brand awareness, and offer free trials
L	L	H	Customers Needing Attention	Low recency, Low frequency but high monetary values.	Make limitedtime offers, and recommend based on past purchases. Reactivate them.
L	L	L	Lost	Lowest recency, frequency, and monetary scores.	Revive interest with reach-out campaign, ignore otherwise.

RFM Model- Assumptions

Finding Recency-

- we took the last invoice date per customer and subtracted today's date from it

Finding Frequency-

- we counted the number of unique invoice numbers per customer.

Finding Monetary Value-

- we multiplied per unit cost to the number of items per invoice and totaled the same for every customer.

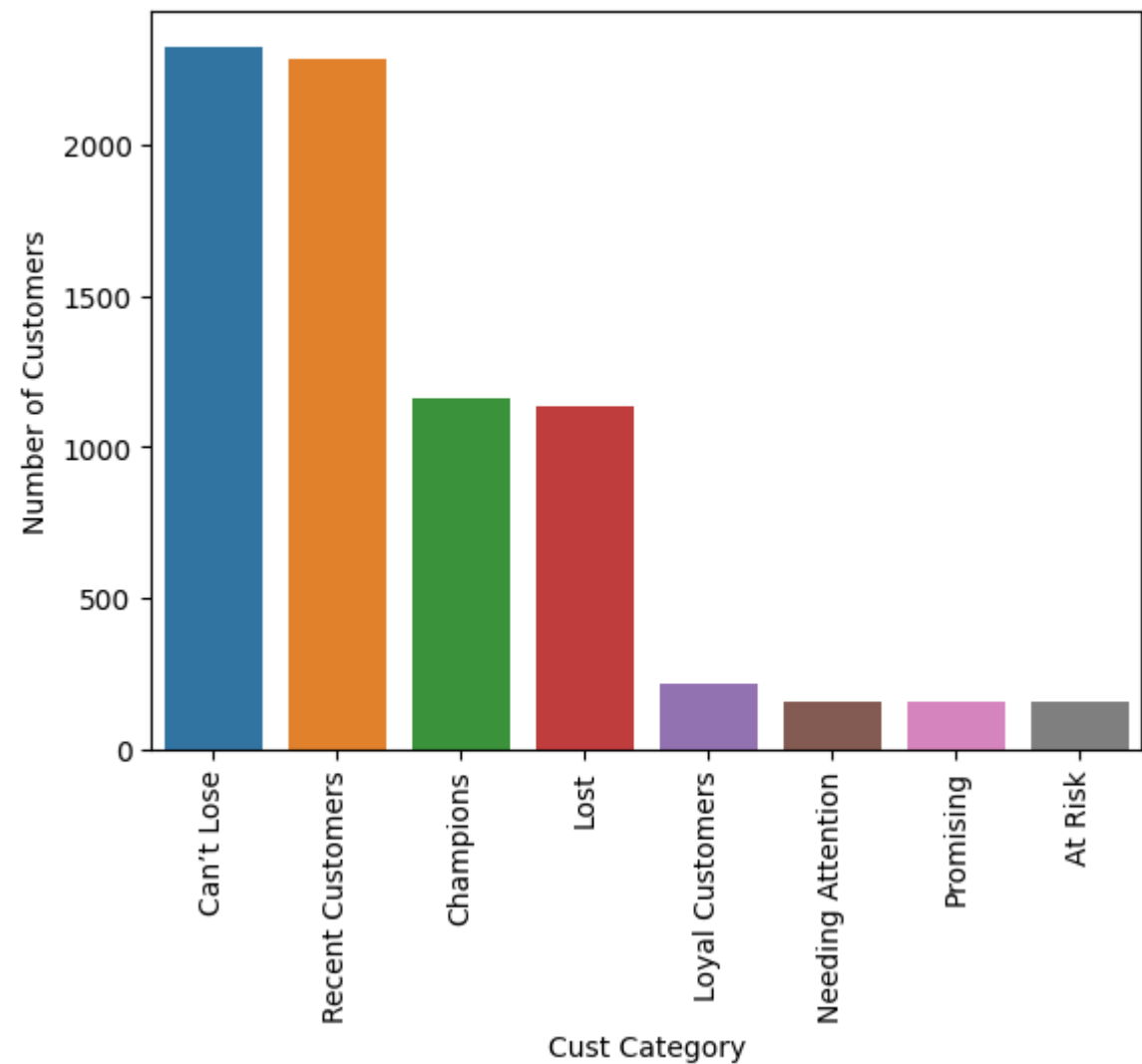
Categorization of Recency, Frequency and Monetary Value as High / Low-

- Since there are only 2 categories- High and Low- We calculated the 50th percentile for each parameter as cutoff.
- All numbers less than cutoff are called L
- All numbers more than the cutoff are called H

Calculation of Final Category of Customer

- Since we have already calculated High/ Low for the three parameters- Recency, Frequency, Monetary Value- we just concatenated these values in a separate column.
- And replaced HHH with "Champions", HHL with "Loyal Customers" .. And so on as per RFM table.

RFM Model- Outcome



RFM Customer Segmentation Result

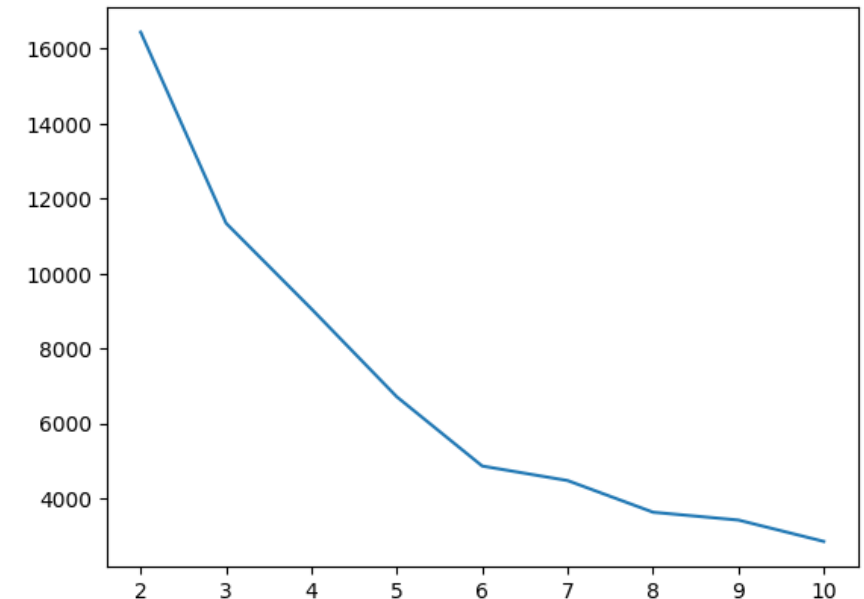
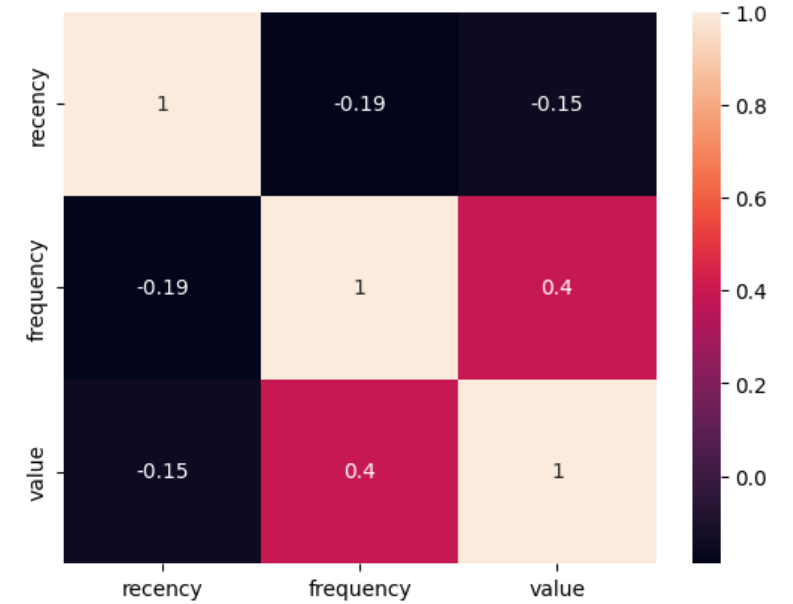
index	cust_category
Can't Lose	2322
Recent Customers	2282
Champions	1158
Lost	1137
Loyal Customers	216
Needing Attention	159
Promising	158
At Risk	155

K Means Model- Assumptions

K Means is an industry known Unsupervised ML Algorithm used for customer classification.

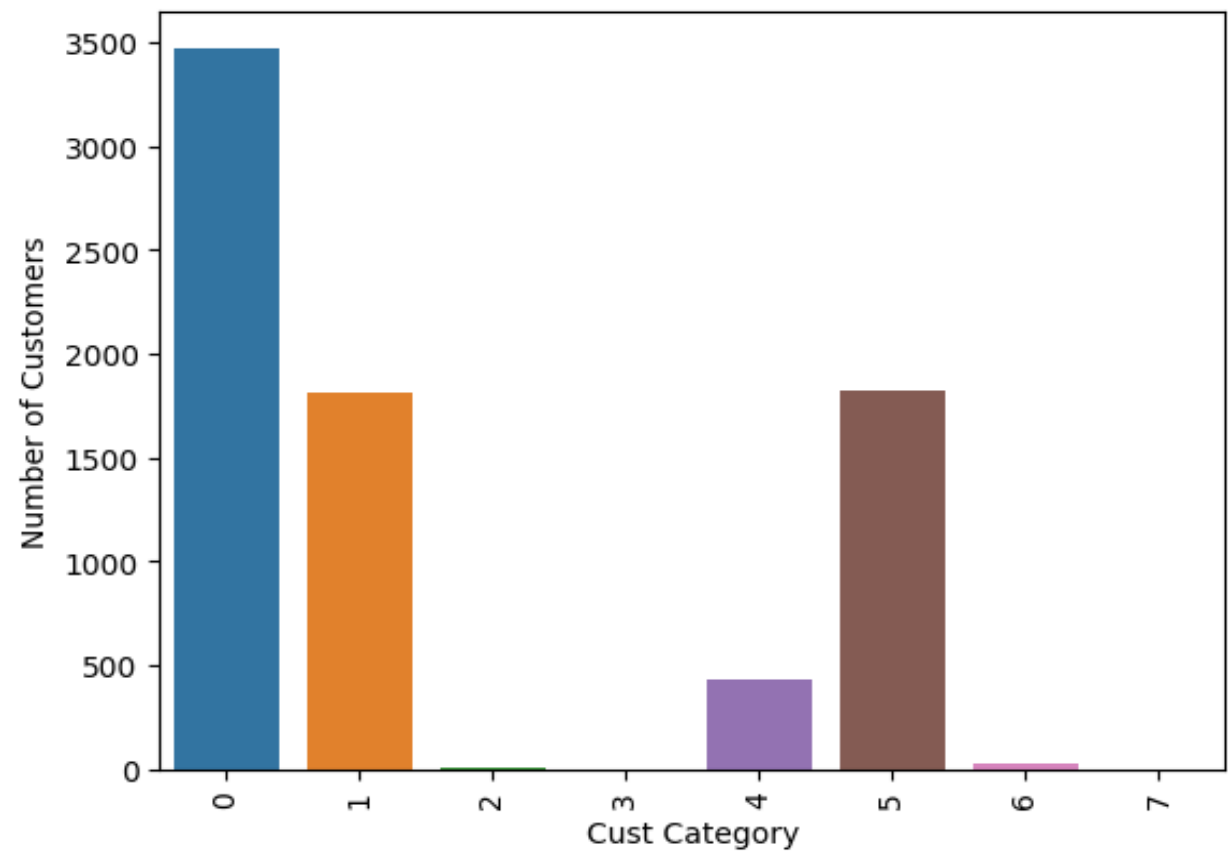
We have used the same 3 customer classification parameters- Recency, Frequency, Monetary Value here also.

- Step1- **Normalization** of Values
- Step2- **Correlation** of Recency, Frequency and Monetary Value with one another.
- Finding appropriate “**elbow curve**” for n_clusters.
- We have taken **n_clusters = 8** (so that outputs of K Means and RFM models can be compared).
- Did model.fit() with n_clusters=8



K Means Model- Output

K Means customer segmentation



index	cust_category
Can't Lose	2322
Recent Customers	2282
Champions	1158
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Loyal Customers	216
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At Risk	155

Model Evaluation & Inferences

K Means & RFM

- We evaluated RFM Model and K Means Machine learning models for customer segmentation.
- We also tried to keep n_clusters=8 for both the models.
- **Both models work similarly with small/ minor differences** (marked in pink color).

Pink Color means diff bet. RFM and K Means results

CustomerID	RFM model cust category	K Means Model Cust category
12346	Recent Customers	1
12747	Can't Lose	0
12748	Can't Lose	7
12749	Can't Lose	4
12820	Can't Lose	0
12821	Recent Customers	5
12822	Can't Lose	0
12823	Needing Attention	0
12824	Can't Lose	0
12826	Can't Lose	0
12827	Can't Lose	0
12828	Can't Lose	0
12829	Recent Customers	1
12830	Can't Lose	0
12831	Recent Customers	1
12832	Can't Lose	0
12833	Champions	5
12834	Champions	1
12836	Can't Lose	0
12837	Recent Customers	5

Future Potential

- The model and its techniques can be extrapolated for other situations, industries, clients, problems for segmentation purposes.

Conclusion

- Both RFM and K Means are useful and power models for customer classification and can be used depending up on business needs.

Thanks